

A disproof of the conjectured recurrence for OEIS A076217

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Abstract

OEIS A076217 carries a conjectured linear recurrence of order three, contributed by Colin Barker. We disprove it, and not merely by exhibiting a counterexample: we prove that the recurrence fails at $n = 3^k$, $n = 3^k + 1$ and $n = 3^k + 2$ for *every* integer $k \geq 2$, so the set of indices at which it fails is infinite. The proof is elementary and self-contained. Everything follows from one fact about the sequence, that $a(n) = 1$ exactly when $n = 3^k - 2$, which is established by an induction driven directly by the entry's own defining recursion; the values at the six indices surrounding each power of three are then computed in closed form, and the failure is read off. The entry records the observation that the recurrence “seems to fail at $n =$ powers of 3”; what is proved here is that it does, always.

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1 The sequence and the conjecture

OEIS A076217 is defined by its name, which is the recursion itself:

$$a(1) = 1, \quad a(n) = a(n-1) + n \cdot \text{sign}(n - a(n-1)) \quad (n \geq 2), \quad (1)$$

where $\text{sign}(x)$ is 1, 0 or -1 according as x is positive, zero or negative. It has offset 1 and begins

$$a(1), \dots, a(12) = 1, 3, 3, 7, 2, 8, 1, 9, 9, 19, 8, 20, \dots$$

Nothing else is taken from the entry: (1) is the definition, so every value used below is computed from it.

The entry carries the following comment:

Conjecture: $a(n) = -a(n-1) + a(n-2) + a(n-3)$ for $n > 5$. G.f.: $-x(27x^{28} + 54x^{27} + 27x^{26} + 9x^{10} + 18x^9 + 9x^8 + 3x^4 + 6x^3 + 5x^2 + 4x + 1)/((x-1)(x+1)^2)$. — Colin Barker, Feb 25 2013

As of the “Last modified” line on the live entry (2025-03-21, revision 33) the statement is still recorded as a conjecture. The entry also carries a later remark that the recurrence “seems to fail at $n =$ powers of 3, and the 2 successive terms”. That is an observation about the terms in range; the theorem below proves it for all k , and the conjectured generating function falls with the recurrence, since a rational function with denominator $(x-1)(x+1)^2$ forces

$$a(n) + a(n-1) - a(n-2) - a(n-3) = 0 \quad (2)$$

for all n past the degree of its numerator.

2 Where the sequence takes the value one

Write $p = 3^k$ throughout. The whole argument rests on one proposition.

Proposition 1. *For every $k \geq 1$, $a(3^k - 2) = 1$.*

The proof is an induction, and the induction step also produces the closed forms we need.

Lemma 2. *Let $p \geq 3$ and suppose $a(p - 2) = 1$. Then*

$$a(p - 1) = p, \quad a(p) = p,$$

and for every i with $0 \leq i \leq p - 2$,

$$a(p + 1 + 2i) = 2p + 1 + i, \quad a(p + 2 + 2i) = p - 1 - i.$$

Proof. All four statements are applications of (1), and in each one the only thing to check is the sign of $n - a(n - 1)$.

At $n = p - 1$ the previous value is $a(p - 2) = 1$ and $n - a(n - 1) = p - 2 > 0$, so the sign is +1 and $a(p - 1) = 1 + (p - 1) = p$.

At $n = p$ the previous value is $a(p - 1) = p$, so $n - a(n - 1) = 0$, the sign is 0, and $a(p) = a(p - 1) = p$.

Now argue by induction on i . For the odd-offset index $n = p + 1 + 2i$ the previous value is $a(p + 2i)$, which is $a(p) = p$ when $i = 0$ and $p - 1 - (i - 1) = p - i$ when $i \geq 1$; in both cases it equals $p - i$ for $i \geq 1$ and p for $i = 0$, and in both cases

$$n - a(n - 1) = (p + 1 + 2i) - (p - i) = 1 + 3i > 0 \quad (i \geq 1),$$

while for $i = 0$ it is $(p + 1) - p = 1 > 0$. So the sign is +1 and

$$a(p + 1 + 2i) = (p - i) + (p + 1 + 2i) = 2p + 1 + i$$

for $i \geq 1$, and $a(p + 1) = p + (p + 1) = 2p + 1$ for $i = 0$, which is the same formula.

For the even-offset index $n = p + 2 + 2i$ the previous value is $a(p + 1 + 2i) = 2p + 1 + i$, and

$$a(n - 1) - n = (2p + 1 + i) - (p + 2 + 2i) = p - 1 - i,$$

which is positive exactly when $i \leq p - 2$. So for those i the sign is -1 and

$$a(p + 2 + 2i) = (2p + 1 + i) - (p + 2 + 2i) = p - 1 - i,$$

as claimed. This completes the induction on i and the lemma. □

Proof of Proposition 1. For $k = 1$ we have $3^1 - 2 = 1$ and $a(1) = 1$ by definition. Suppose $a(3^k - 2) = 1$ for some $k \geq 1$ and put $p = 3^k \geq 3$. Lemma 2 applies. Take $i = p - 2$, which is within the permitted range $0 \leq i \leq p - 2$; then

$$p + 2 + 2i = p + 2 + 2(p - 2) = 3p - 2 = 3^{k+1} - 2,$$

and the lemma gives

$$a(3^{k+1} - 2) = p - 1 - i = p - 1 - (p - 2) = 1.$$

That is the statement for $k + 1$. □

Corollary 3. *For every $k \geq 1$, writing $p = 3^k$,*

$$a(p - 3) = p - 1, \quad a(p - 2) = 1, \quad a(p - 1) = p, \quad a(p) = p, \quad a(p + 1) = 2p + 1, \quad a(p + 2) = p - 1.$$

Proof. The values at $p-2$, $p-1$, p , $p+1$ and $p+2$ are Proposition 1 together with Lemma 2 at $i=0$.

For $a(p-3)$, apply Lemma 2 at the previous power. Put $q = 3^{k-1}$, so $p = 3q$ and $p-3 = 3q-3$. The lemma describes the indices after q in two families, $q+1+2i$ and $q+2+2i$. Solving $q+2+2i = 3q-3$ gives $2i = 2q-5$, which has no integer solution, so $p-3$ is not of the second kind. Solving $q+1+2i = 3q-3$ gives

$$i = q - 2,$$

which lies in the permitted range $0 \leq i \leq q-2$, and the first family gives

$$a(p-3) = 2q+1+i = 2q+1+(q-2) = 3q-1 = p-1.$$

This is legitimate for $k \geq 2$, since then $q \geq 3$ and Lemma 2 applies at q ; and $i = q-2$ is the same index at which the lemma yields $a(3q-2) = 1$, so both values come from a single application. $\square$

3 The recurrence fails at every power of three

Theorem 4. *For every integer $k \geq 2$ the conjectured recurrence (2) fails at $n = 3^k$, at $n = 3^k+1$ and at $n = 3^k+2$. In particular it fails at infinitely many indices, so it does not hold for all $n > 5$, and the conjectured generating function is likewise incorrect.*

Proof. Put $p = 3^k$ and use Corollary 3.

At $n = p$ the right-hand side of the conjecture is

$$-a(p-1) + a(p-2) + a(p-3) = -p+1+(p-1) = 0,$$

whereas $a(p) = p$. Since $p = 3^k \geq 9 > 0$, the two differ.

At $n = p+1$ it is

$$-a(p) + a(p-1) + a(p-2) = -p+p+1 = 1,$$

whereas $a(p+1) = 2p+1 \geq 19$.

At $n = p+2$ it is

$$-a(p+1) + a(p) + a(p-1) = -(2p+1) + p + p = -1,$$

whereas $a(p+2) = p-1 \geq 8$.

All three indices exceed 5, so each is inside the range the conjecture claims. As k ranges over the integers ≥ 2 these give infinitely many failures. $\square$

4 Verification

Two checks were made, independent of the algebra above.

First, the recursion (1) was iterated in exact integer arithmetic to $n = 60000$ and the resulting values compared with the entry's DATA field and with its b-file, which carries 10000 terms; they agree at every index.

Second, the conjectured recurrence was evaluated on those values at every index from 6 to 4000. It fails at exactly the indices

$$9, 10, 11, \quad 27, 28, 29, \quad 81, 82, 83, \quad 243, 244, 245, \quad 729, 730, 731, \quad 2187, 2188, 2189,$$

that is, at $3^k, 3^k+1, 3^k+2$ for $k = 2, \dots, 7$ and nowhere else, matching the theorem exactly. The closed forms of Corollary 3 were checked at every $k \leq 9$ and the block formulas of Lemma 2 at every admissible i in that range.

5 Concluding remarks

The conjecture is not close to true: it fails on a set of indices of the shape $3^k + \{0, 1, 2\}$, and between consecutive powers of three it holds only because the sequence is locked into the simple alternating pattern of Lemma 2, which the recurrence happens to reproduce. The pattern is broken precisely at the moment the downward branch reaches the value 1, which by Proposition 1 happens exactly at $n = 3^k - 2$.

References

- [1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry [A076217](#), 2025.